

Std - X

Sub - Science



CHAPTER 1

Chemical Reactions and Equations





CHAPTER 1

Chemical Reactions and Equations

Part – 4

Types of Chemical Reactions

B) Decomposition Reaction

A chemical reaction in which 'a single reactant' breaks down to give 'two or more simpler products' is called decomposition reaction.



B) Decomposition Reaction

**Thermal
Decomposition**

**Electrolytic
Decomposition**

**Photolytic
Decomposition**

B) Decomposition Reaction

A chemical reaction in which 'a single reactant' breaks down to give 'two or more simpler products' is called decomposition reaction.

ACTIVITY 1.5 - Decomposition of Ferrous Sulphate

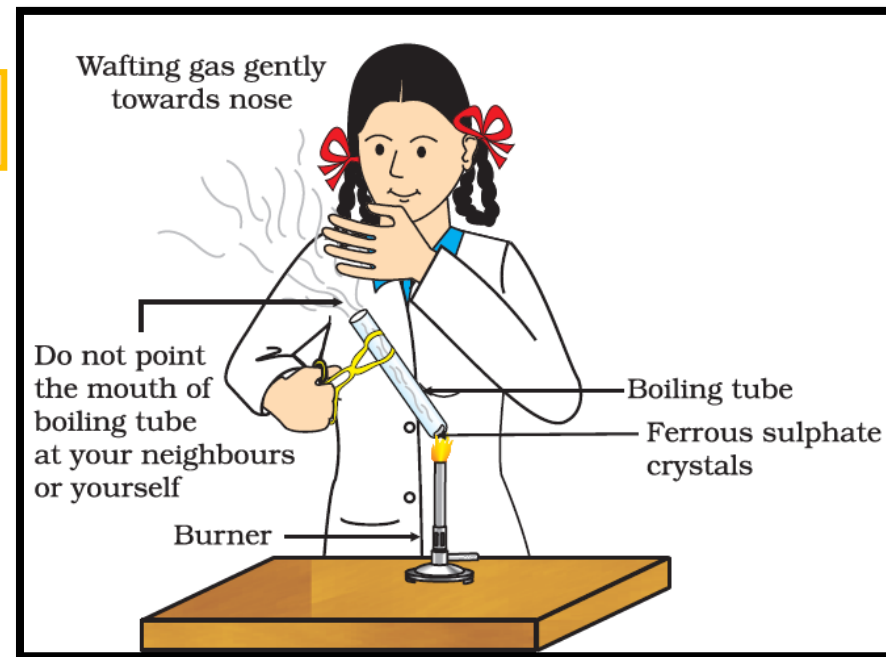
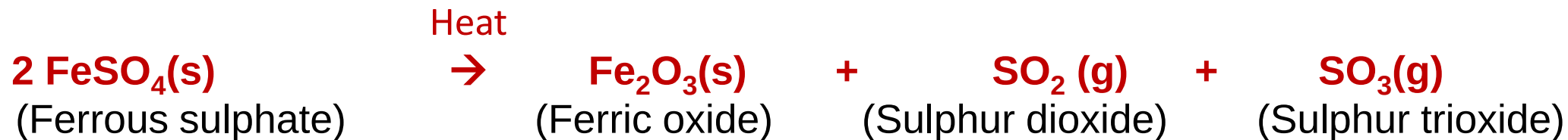
Be Careful !

Do not point the mouth of the boiling tube at your neighbours or yourself.

Example

When ferrous sulphate crystals are heated in a boiling tube, it decomposes to give 'ferric oxide along with sulphur dioxide and sulphur trioxide gases.

Chemical equation:



B) Decomposition Reaction

Chemical equation :

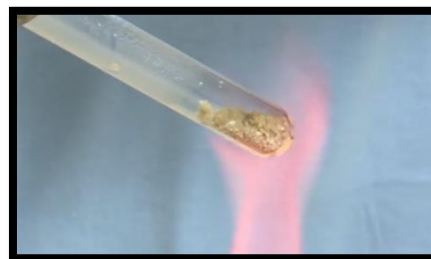


$2 \text{FeSO}_4(\text{s})$
(Ferrous sulphate)
(Light Green)



Heat

→



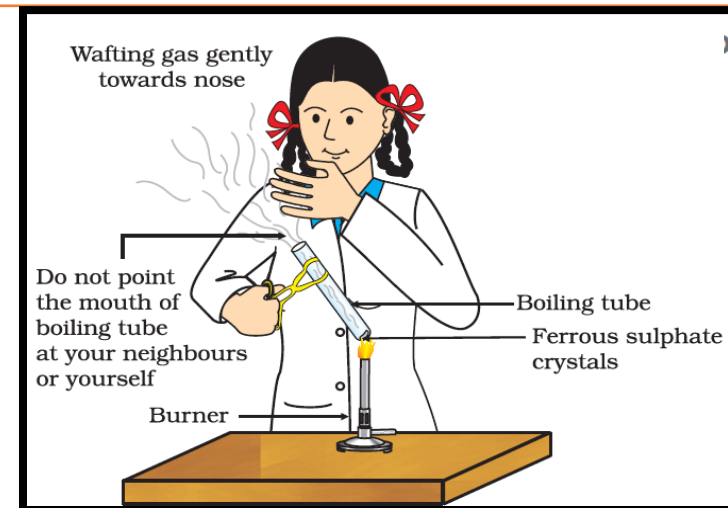
$\text{Fe}_2\text{O}_3(\text{s})$
(Ferric oxide)
(White)

+

$\text{SO}_2(\text{g})$
(Sulphur dioxide)

+

$\text{SO}_3(\text{g})$
(Sulphur trioxide)

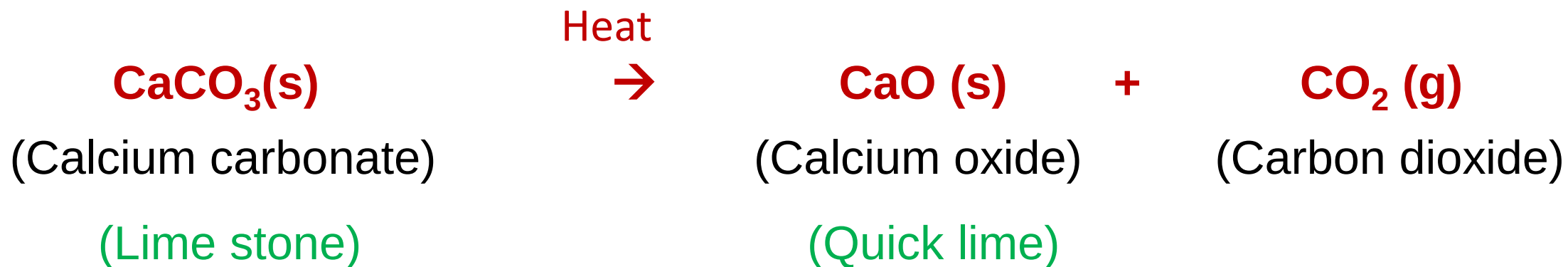


- ✓ In this reaction you can observe that a single reactant breaks down to give simpler products.
- ✓ This is a decomposition reaction.
- ✓ Ferrous sulphate crystals ($\text{FeSO}_4, 7\text{H}_2\text{O}$) lose water when heated and the colour of the crystals changes.
- ✓ It then decomposes to ferric oxide (Fe_2O_3), sulphur dioxide (SO_2) and sulphur trioxide (SO_3).
- ✓ Ferric oxide is a solid, while SO_2 and SO_3 are gases.

B) Decomposition Reaction

- Thermal Decomposition –
- When a decomposition reaction is carried out by heating, it is called thermal decomposition.

Chemical equation :



B) Decomposition Reaction

- Thermal Decomposition –

- Decomposition of calcium carbonate to calcium oxide and carbon dioxide on heating is an important decomposition reaction used in various industries.
- Calcium oxide is called lime or quick lime.



$\text{CaCO}_3(\text{s})$
(Calcium carbonate)
(Lime stone)

Heat
 $\rightarrow$



$\text{CaO}(\text{s})$
(Calcium oxide)
(Quick lime)



$\text{CO}_2(\text{g})$
(Carbon dioxide)
(turns lime water milky)

- It has many uses – one is in the manufacture of cement.

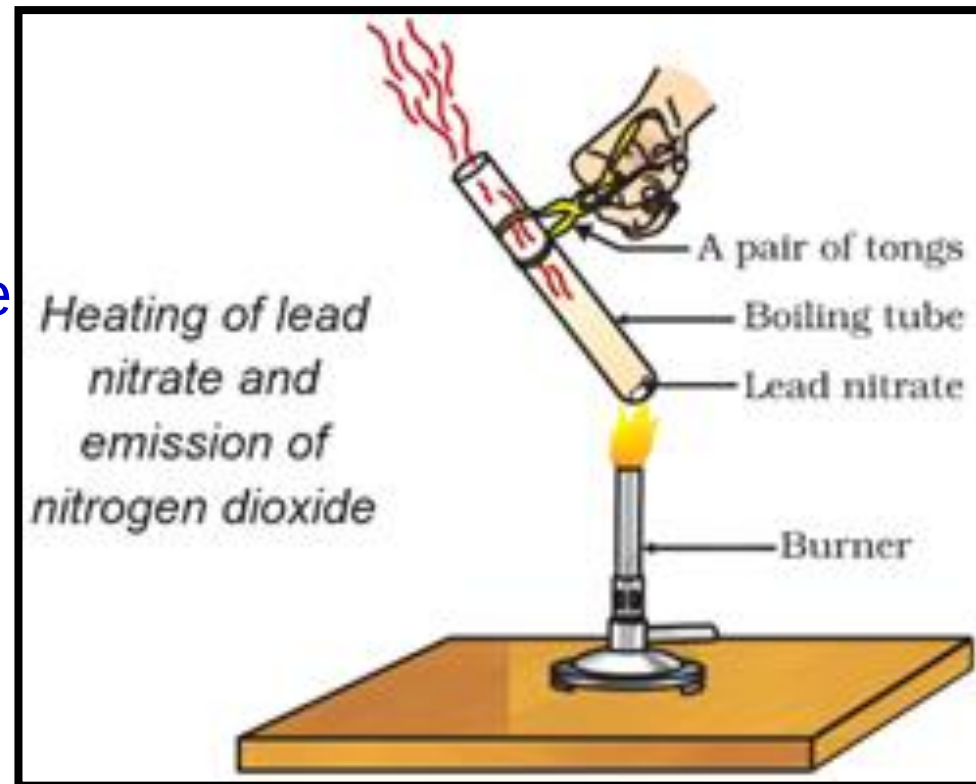
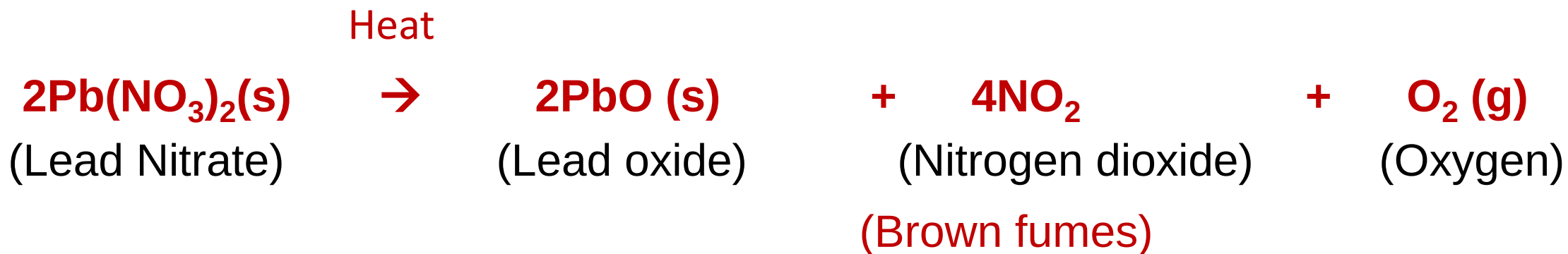
B) Decomposition Reaction

- Thermal Decomposition –

Activity 1.6 – Decomposition of Lead Nitrate

- When Lead nitrate powder is heated in a boiling tube it decomposes to form lead oxide, nitrogen dioxide (brown fumes) and oxygen gas.

Chemical equation :

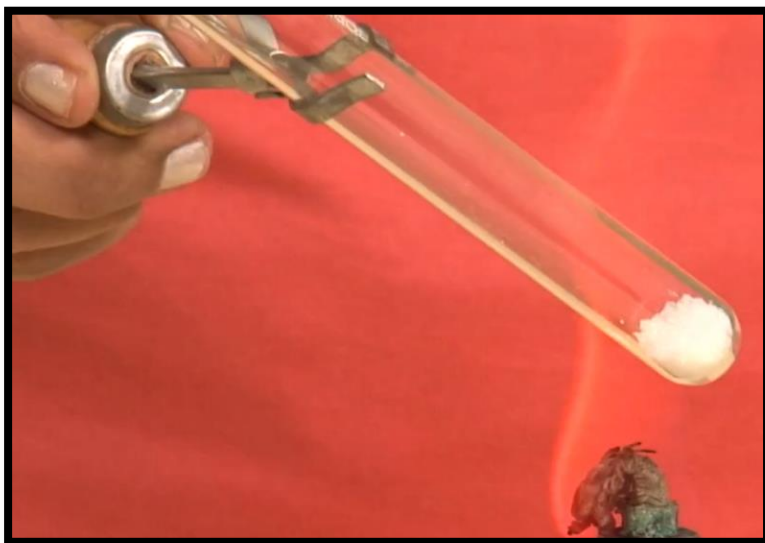


B) Decomposition Reaction

- Thermal Decomposition –

Activity 1.6 – Decomposition of Lead Nitrate

Chemical equation



Heat

$2\text{Pb}(\text{NO}_3)_2(\text{s})$
(Lead Nitrate)

→

$2\text{PbO}(\text{s})$
(Lead oxide)

+

4NO_2
(Nitrogen dioxide)
(Brown fumes)

+

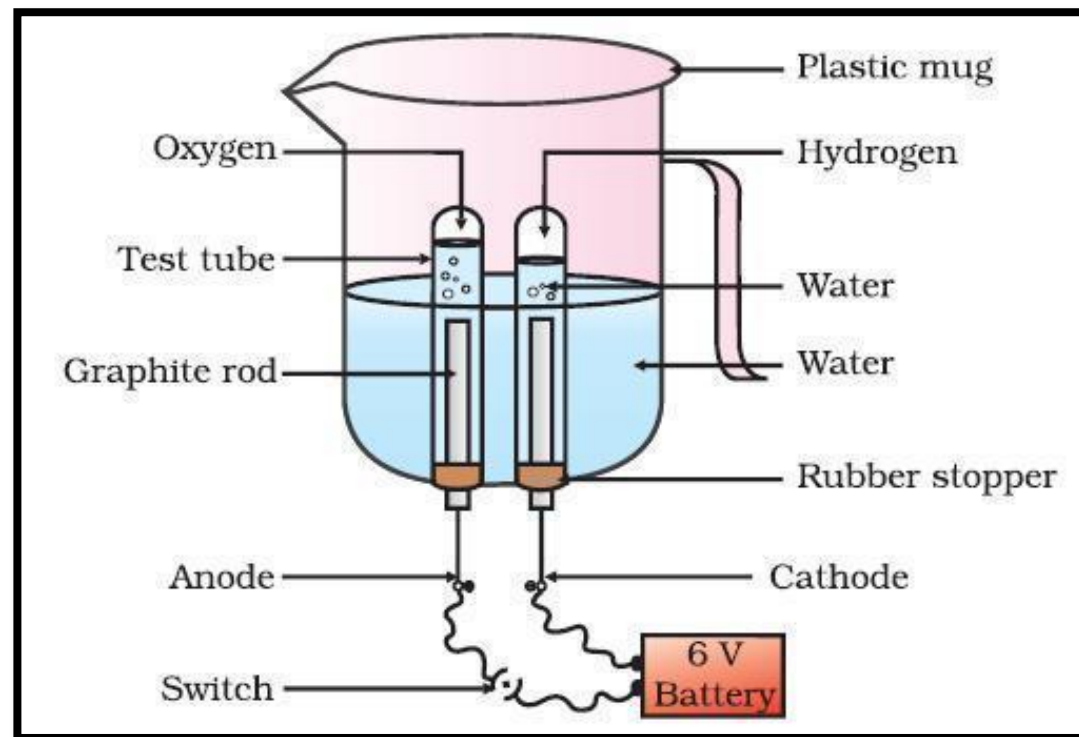
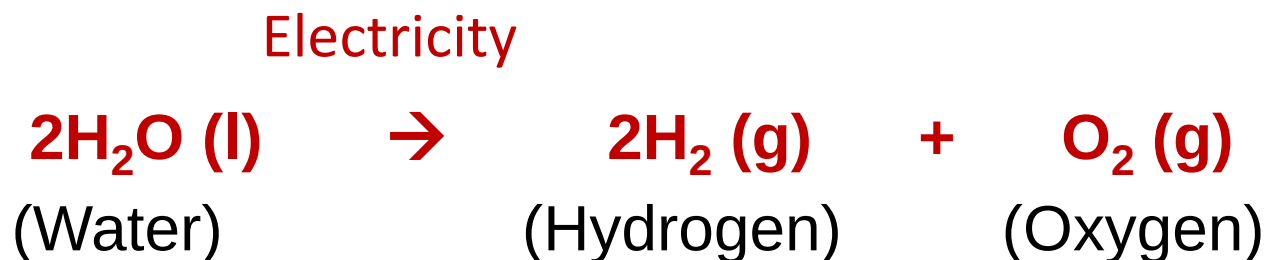
$\text{O}_2(\text{g})$
(Oxygen)



B) Decomposition Reaction

- Electrolytic Decomposition –
- When a decomposition reaction is carried out by passing electric current through aqueous solution, it is called electrolytic decomposition.

Activity 1.7 – Electrolysis of water



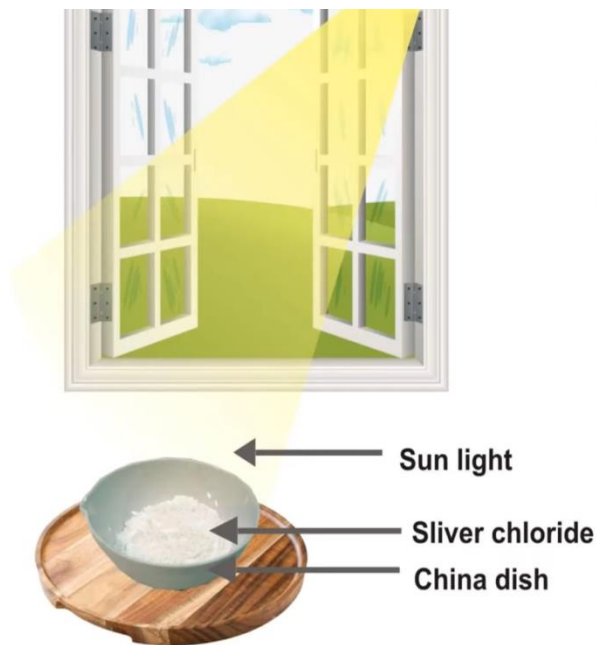
B) Decomposition Reaction

- Photo Decomposition – **Activity 1.8 – Photodecomposition of Silver Chloride**

When a decomposition reaction is carried out in presence of light, it is called photo decomposition reaction.



2AgCl (s)
(Silver Chloride)



Sunlight
→



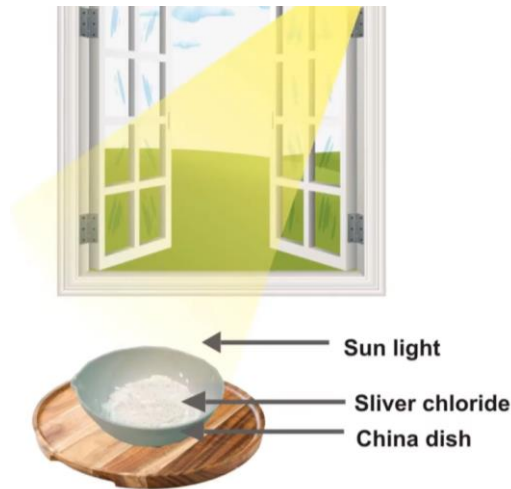
2Ag (s) + **$\text{Cl}_2 \text{(g)}$**
(Silver) (Chlorine)

B) Decomposition Reaction

- **Photo Decomposition** – Activity – 1.8 – Photo-decomposition of Silver Chloride
- White silver chloride turns grey in sunlight.
- This is due to the decomposition of silver chloride into silver and chlorine by sunlight.



2AgCl (s)
(Silver Chloride)



Sunlight
→



2Ag (s) + **$\text{Cl}_2 \text{(g)}$**
(Silver) (Chlorine)

B) Decomposition Reaction

- **Photo Decomposition** – Photo-decomposition of Silver Bromide
- White silver bromide turns grey in sunlight.
- This is due to the decomposition of silver bromide into silver and bromine by sunlight.



2AgBr (s)
(Silver Bromide)

Sunlight



2Ag (s) + **$\text{Br}_2 \text{(g)}$**
(Silver) (Bromine)

B) Decomposition Reaction

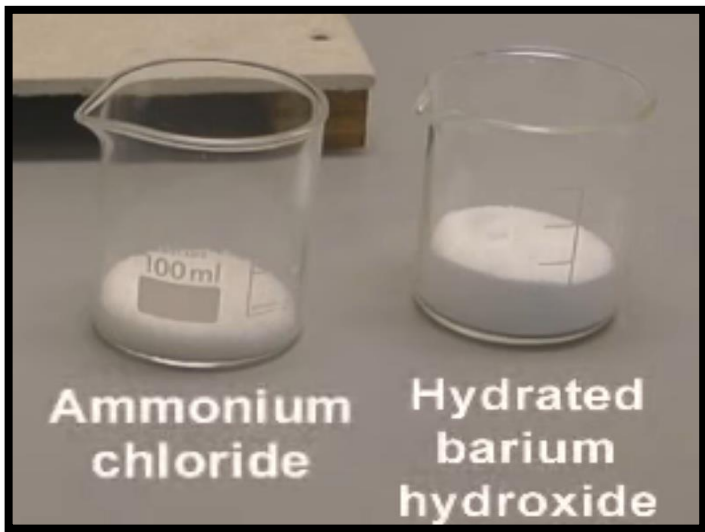
- Photo Decomposition – Photodecomposition of Silver Bromide



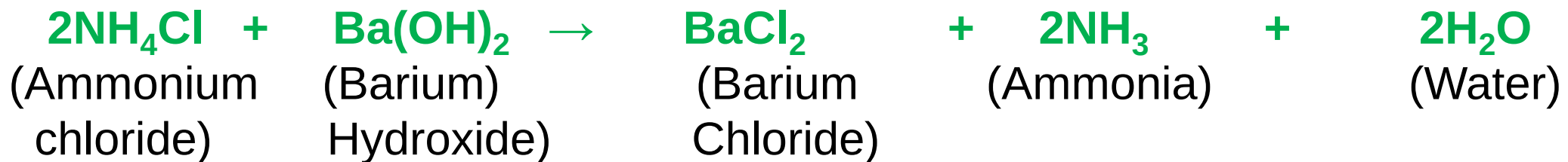
Photodecomposition of silver bromide is used in 'Black and White Photography'.

B) Decomposition Reaction

- Endothermic Reaction –
- The decomposition reactions require energy either in the form of heat, light or electricity for breaking down the reactants.
- Reactions in which energy is absorbed are known as endothermic reactions.



The beaker becomes very cold. This is due to endothermic reaction.





Home - Work



Q U E S T I O N S

1. A solution of a substance 'X' is used for white washing.
 - (i) Name the substance 'X' and write its formula.
 - (ii) Write the reaction of the substance 'X' named in (i) above with water.
2. Why is the amount of gas collected in one of the test tubes in Activity 1.7 double of the amount collected in the other? Name this gas.





Home - Work

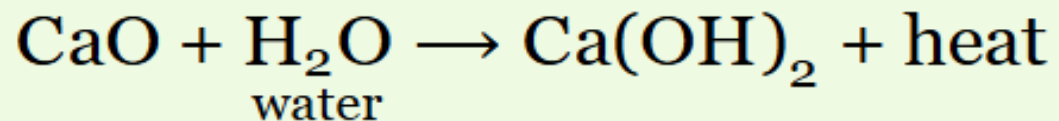


Q U E S T I O N S

1. A solution of a substance 'X' is used for white washing.
- (i) Name the substance 'X' and write its formula.
 - (ii) Write the reaction of the substance 'X' named in (i) above with water.

(i) The name of the substance used for whitewashing is calcium oxide and its formula is CaO.

(ii) It's reaction with water is-





Home - Work

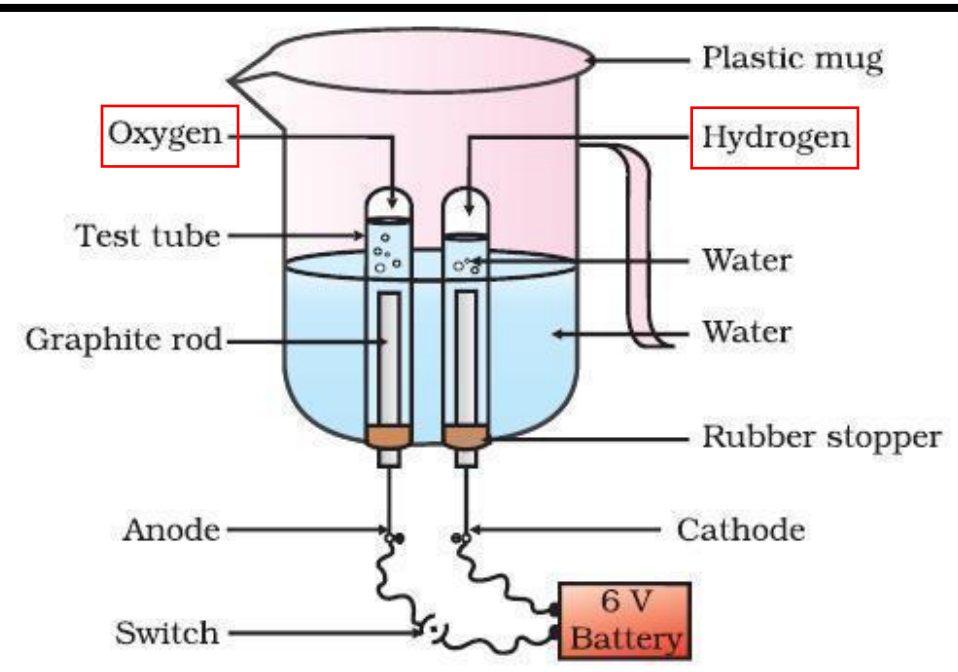


Q U E S T I O N S

2. Why is the amount of gas collected in one of the test tubes in Activity 1.7 double of the amount collected in the other? Name this gas.



- Water (H_2O) contains two parts of hydrogen and one part oxygen.
- Therefore, the amount of hydrogen and oxygen produced during electrolysis of water is in a 2:1 ratio.
- During electrolysis, hydrogen gas is collected in one test tube and oxygen is collected in another test tube.
- Thus, the amount of gas collected in one of the test tubes is double of the amount collected in the other.



TO BE CONTINUED

